Daily Drilling Report
SPD19B-03
4/15/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
21.0	16"	42.00	72.83	31.70	103.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	216,898.34	790.30	4,884,251.55		18.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,492.00	1,492.00	0.00			

Operations at Report Time
RIH 13 3/8" casing @ 1430m.

Operations Summary
POOH directional BHA#6 f/ 750m to surface. Retrieve wear bushing. Wash inside WH area. R/U 13 3/8" casing handling tools. RIH 13 3/8" casing to 1030m.

Operations next 24hrs
Cont' RIH 13 3/8" casing and Land on CHH. Perform cement job. N/D 21 1/4" BOP. Set pack-off. N/U Unihead wellhead. Skid over well SPD19B-04.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/29/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	4/12/2014	6/2/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	4/14/2014	6/14/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill	4/9/2014	5/25/2014
Trip Drill	4/8/2014	5/29/2014

STOP Cards

Number of Reports: 31

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	04:30	4.50	16"	D	04	P	POOH directional BHA#6 f/ 750m to surface. L/D 8" Jar. R/B BHA. Check the bit (1-1-WT-G-E-I-NO-TD). ** Motor bearing gap: 1mm
04:30	05:00	0.50	16"	C	05	P	P/U & M/U Aker Wear Bushing R/T to one std of 5 7/8" DP, RIH to 19.45m, retrieve WB, pull string to surface, R/B std.
05:00	05:30	0.50	16"	G	20	P	P/U & M/U & RIH Aker washing tool and wash inside BOP & WH w/ 5 rpm/ 550 gpm/ 170 psi. POOH to surface and L/D same.
05:30	06:30	1.00	16"	G	01	P	L/D drilling Bail. M/U 5 ft pup joint + XOS + AL-Ghaith Fill up tools to TDS. Install 22' long bails. R/U Side door elevator + single joint elevator + Power tong + manual back up tong.
06:30	07:00	0.50	16"	G	24	N	(NPT for AL-Ghaith Company) Delay in operation due to problem to R/U Power tong.
07:00	07:30	0.50	16"	G	18	P	HPJSM. P/U 13 3/8" casing shoe joint (68 ppf, BTC, L-80), Botil float shoe already made up. Check float passing through and holding, OK. P/U and M/U 1 joint 13 3/8" casing + Float collar joint (Float collar already made up), check float collar passing through, OK. RIH to 36m. ** Apply baker lock on all shoe track connections. ** Install 2 Bow centralizer on shoe joint. ** Install 2 Bow centralizer on intermediate joint. ** Install 1 Bow centralizer on float collar joint.
07:30	08:00	0.50	16"	G	24	N	(NPT for AL-Ghaith Company) AL-Ghaith power tong could not apply 16000 TRQ to M/U shoe track joints with baker lock. It could not hold the pipe and rotate it. Had to try several times to apply this amount of TRQ caused un-normal scratches on shoe track joints bodies.
08:00	09:00	1.00	16"	G	18	P	Cont' run 13 3/8" CSG (68 ppf, L-80, BTC) from 36m to 200m.
09:00	11:00	2.00	16"	G	24	N	(NPT for AL-Ghaith Company) Had severe problems with AL-Ghaith power tongs. It could not apply enough torque to M/U casing joints and make deep scratches on casing body, it could not hold the pipe and rotate it. Had to change tong dices several times, still powr tong spinning with very low speed consequently taking longer time to make connections and scratching the casings bodies. >>> As per Petropars Experience in RIH 13 3/8" casing in cased hole well and poor performance of AL-Ghaith power tong (that lead to severe delay in operation), 2 hrs downtime considered for AL-Ghaith company for this interval.
11:00	11:30	0.50	16"	G	01	P	Remove side door elevator. R/U spider elevator & FMS.

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
11:30	15:00	3.50	16"	G	18	P	Cont' run 13 3/8" CSG f/ 200m to 870m (18 5/8" CSG shoe). ** Install bow centralizers ar per tally on stop collars.
15:00	20:30	5.50	16"	G	24	N	(NPT for AL-Ghaith Company) Had continuous problem with power tong. It could not hold casing joints and apply torque. Change tong to back up one, same problem. Change and modify the dices several times, no result. Power tong spining with very low speed and taking longer connection time. Power tong applying required TRQ with difficulty and causing scratches on casing body in result. >> As per previous experiences in RIH 13 3/8" casing in cased hole wells with FMS and spider elevator, 5.5 hrs downtime considered for AL-Ghaith casing company. >> Note: AL-Ghaith FMS could not work properly. Closing by itself. >> AL-Ghaith company did not send enough and proper power tong dices for job. >> AL-Ghaith company sent very poor quality of tools.
20:30	21:30	1.00	16"	G	07	P	Circulate 500 bbls (1.2 casing volume) w/ 450 gpm/ 150 psi.
21:30	22:30	1.00	16"	G	18	P	Cont' run 13 3/8" CSG f/ 870m to 1030m. ** Install bow centralizers ar per tally on stop collars. ** Fill up each joint while RIH.
22:30	00:00	1.50	16"	G	24	N	(NPT for AL-Ghaith Company) Due to poor equipment performance of AL-Ghaith (power tong), 1.5 hr downtime considered for this interval.
00:00	00:00						Apr 16 , 2014 00:00 - 02:00 Cont' run 13 3/8" CSG f/ 1030m to 1430m. ** Install bow centralizers ar per tally on stop collars. ** Fill up each joint while RIH. 02:00 - 06:00 (NPT for AL-Ghaith Company) Due to poor equipment performance of AL-Ghaith (power tong), 4 hr downtime considered for this interval.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,492.00	Polymer Mud	9.20	43	12.0	20.000	8.8
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
8.8						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	6	0.0	93.0	7.0	24,200.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	6.000		0/100		1755.0	2037.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
2748.0	0.0	0.0			13.09539	0.409

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	5.	10	6.0	1,007.0	32.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Navis King	Supply Boat	4/14/2014 01:00		Stdby Anchorage.
Ahmed-6	Supply Boat	4/14/2014 12:00	4/15/2014 17:15	Delivered Cargo from Sahar 2, B/Loaded Cargo for Kish by snatch, Dep for Kish.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk	3.0	216.0	70.0
H2S Scavenger	4.0	120.0	41.0
Barite Bulk	6.0	162.0	149.0
Lime	10.0	338.0	107.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	15.0
Cement G	0.0	0.0	168.0
Bentonite BB	0.0	0.0	21.0
Bentonite Bulk	3.0	0.0	62.0
Barite Bulk	6.0	0.0	141.0
Diesel Fuel	7.0	0.0	103.0
Potable Water	20.0	20.0	110.0
Drill Water	70.0	0.0	358.0

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Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
6	16	Varel	CR14JMRS	1244888	Insert	435	20/20/20/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.11	6.68	61,723.20	1,478.00	1,492.00	14.00	2.1	1-1-WT-G-E-I-NO-TD
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
800	800	1,800.0	1,800.0	42	42	154	154
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
9.96	6.68	100.00	0.00	349	311	269	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	139.60	313.30
XOS	8	3.000	0.79	173.70
6 jts 8 1/4" DC	8 1/4	2.810	56.24	172.91
8" Drilling Jar	8	3.500	5.72	116.67
8 jts 8 1/4" DC	8 1/4	2.810	75.34	110.95
XOS	9 1/2	3.000	1.22	35.61
9 1/2" Float Sub	9 1/2	3.000	0.78	34.39
9 1/2" NMDC	9 1/2	4.000	9.44	33.61
9 1/2" TM - Sub	9 1/2		2.11	24.17
9 1/2" NMDC	9 1/2	4.000	9.44	22.06
9 1/2" Restrictor sub	9 1/2	4.310	0.92	12.62
14 3/4" Stabilizer	14 3/4	4.000	2.16	11.70
9 5/8" Motor (BH: 1.25, Sleeve: 15 3/4")	9 5/8		9.14	9.54
16" Insert Bit	16		0.40	0.40

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,176.98
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.87
Hith		1,469.30	1,402.67

General Remarks

Comment

>>> Requirements: (20 3/4" Dovetail for BOP Test Plug) + (8" Drilling Jar)

>>> Sent:

** Borkan boat to Kish base: (PDF chemicals: 10 pallet Guar Gum)

** Ahmed-6 boat to Kish base: (1 no 8" Cougar Drilling Jar, COR2898AD)

>>> Received:

** Ahmed-6 f/ rig Sahar-2: (1 no 8" HM Drilling jar)